Deep Learning Systems Analysis Report

RentShield Canada: NLP Detection of Risky Rental Listings

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Report Overview

This project focused on detecting renter-risk language in Toronto rental listings using deep learning. The dataset used was Toronto Kijiji Rental Data from Kaggle, which contains real rental advertisements with listing descriptions and housing details. A Transformer-based natural language processing model was implemented using DistilBERT for binary text classification. The goal was to classify listings into Safe or Risk categories based on rental listing language.

Dataset and Task Description

The dataset contains 4,106 rental listings and 23 columns. Important variables include Title, Price, Bedrooms, Bathrooms, Address, and Description. The most important text field for this project was Description because it contains the main rental advertisement language.

After removing rows with missing descriptions and duplicate text entries, 2,796 usable rows remained. A rule-assisted expanded labeling process was then applied to create a substantially larger binary classification dataset. The final modeling dataset used 2,793 labeled rows:

- Safe = 2,497
- Risk = 296

This substantially expanded dataset was used to address the earlier limitation of having too few labeled samples for reliable deep learning training and evaluation.

This dataset supports a supervised deep learning text classification task.
Dataset Source: Toronto Kijiji Rental Data (Kaggle)
https://www.kaggle.com/datasets/umeshkhatiwada/toronto-kijiji-rental-data

Model Architecture and Design Decisions

The baseline model used DistilBERT with a sequence classification head. DistilBERT is a compressed version of BERT that keeps strong language understanding performance while reducing model size and computational cost (Sanh et al., 2019).

DistilBERT was selected because:
- It is appropriate for text classification tasks
- It trains faster than full BERT
- It performs well on smaller hardware environments
- It is suitable for academic prototype work

The model output layer predicted two classes:
- Safe
- Risk

The AdamW optimizer was used with a learning rate of 2e-5. Fine-tuning pretrained Transformer models is a standard practice in NLP tasks (Devlin et al., 2019).

Experimental Comparison

One major aspect was changed in the experimental model: the loss function was modified using class weights.

Because the dataset had more Safe than Risk examples, weighted cross-entropy loss was used to give additional penalty when Risk samples were misclassified. This was selected to improve minority-class detection.

Baseline model:

- Standard loss

Experimental model:

- Weighted loss favoring Risk class

This was a meaningful comparison because class imbalance is common in real classification problems.

Results and Interpretation

The baseline DistilBERT model achieved 97.14% test accuracy. The weighted-loss experimental model achieved 97.32% test accuracy.

The experimental model also improved minority Risk recall from 0.78 to 0.88, indicating better detection of risky listings. This is important because the Risk category is the more valuable class for practical screening.

Both models performed strongly overall, but the weighted-loss configuration provided the best balance between overall accuracy and minority-class detection.

Limitations and Risks

This project had several limitations:

- The expanded labeled dataset was created using rule-assisted labeling rather than full manual annotation.
- Human labeling decisions may include subjectivity.
- Binary labels simplify real rental risk complexity.
- Listings from one city may reduce generalization to other regions.
- Additional external validation would strengthen confidence.

Although results were strong, additional external datasets would improve robustness.

Ethical and Responsible Use

One important ethical concern is false positive predictions. A legitimate rental listing could be incorrectly flagged as risky, which may unfairly affect landlords or property managers.

Another concern is fairness bias. Some listings containing demographic restrictions or wording patterns may reflect broader housing discrimination issues.

AI systems must be carefully reviewed to avoid reinforcing bias (Barocas et al., 2019).

The model should be used only as decision support, not as an automatic enforcement system.

Future Improvements

With more time and resources, the system could be improved by:

- Creating a larger fully human-validated labeled dataset
- Adding multi-class labels (Safe, Suspicious, High Risk)
- Using cross-validation for stronger evaluation
- Testing larger Transformer models such as BERT or RoBERTa
- Using explainable AI tools such as SHAP or LIME
- Expanding to other Canadian cities

These steps would improve robustness and practical value.

References

Barocas, S., Hardt, M., & Narayanan, A. (2019). *Fairness and machine learning*. fairmlbook.org.

Devlin, J., Chang, M. W., Lee, K., & Toutanova, K. (2019). BERT: Pre-training of deep bidirectional transformers for language understanding. *NAACL-HLT*.

Sanh, V., Debut, L., Chaumond, J., & Wolf, T. (2019). DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter. arXiv:1910.01108.